Reaction Mechanisms and Catalysis

Answer key with worked solutions

A. Elementary steps

1. Molecularity.

> (a) unimolecular: rate $= k[\mathrm{O_3}]$ (b) bimolecular: rate $= k[\mathrm{NO}][\mathrm{O_3}]$ (c) bimolecular: rate $= k[\mathrm{NO_2}]^2$

2. Hydrogen peroxide and iodide.

> (a) Adding the steps and cancelling $\mathrm{I^-}$ and $\mathrm{IO^-}$: $2\,\mathrm{H_2O_2} \longrightarrow 2\,\mathrm{H_2O} + \mathrm{O_2}$. (b) $\mathrm{I^-}$ is the **catalyst** (used in step 1, regenerated in step 2); $\mathrm{IO^-}$ is the **intermediate** (made in step 1, used in step 2). (c) From the slow step: rate $= k[\mathrm{H_2O_2}][\mathrm{I^-}]$. (d) **Homogeneous**: the catalyst is in the same phase (aqueous solution) as the reactant.

B. Testing mechanisms

3. Two mechanisms for NO + O2.

> (a) A: one elementary step $2\,\mathrm{NO} + \mathrm{O_2} \longrightarrow 2\,\mathrm{NO_2}$, so rate $= k[\mathrm{NO}]^2[\mathrm{O_2}]$. B: slow step rate $= k_2[\mathrm{N_2O_2}][\mathrm{O_2}]$; the pre-equilibrium gives $[\mathrm{N_2O_2}] = K_1[\mathrm{NO}]^2$, so rate $= k_2K_1[\mathrm{NO}]^2[\mathrm{O_2}]$. Both match.
>
> (b) **B**: it uses only bimolecular steps, whereas A needs three molecules to collide at once, which is very unlikely. The rate law alone can't decide between them: it can rule a mechanism out, but not prove one.

4. Ozone.

> Slow step: rate $= k_2[\mathrm{O}][\mathrm{O_3}]$. From the fast equilibrium:
>
> $$K_1 = \frac{[\mathrm{O_2}][\mathrm{O}]}{[\mathrm{O_3}]}, \quad [\mathrm{O}] = \frac{K_1[\mathrm{O_3}]}{[\mathrm{O_2}]}$$
>
> $$\text{rate} = \frac{k_2K_1[\mathrm{O_3}]^2}{[\mathrm{O_2}]} = \frac{k[\mathrm{O_3}]^2}{[\mathrm{O_2}]}$$
>
> Adding $\mathrm{O_2}$ shifts the fast equilibrium to the left, lowering [O], the intermediate needed in the slow step: the reaction is order −1 in $\mathrm{O_2}$.

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C. Catalysts and enzymes

5. Hydrogenation on nickel.

> (1) **Adsorption**: H_2 and ethene bond to active sites on the nickel surface. (2) **Weakening**: the H–H bond (and the C=C π bond) weakens or breaks. (3) **Reaction**: H atoms add to the adsorbed ethene, forming ethane. (4) **Desorption**: ethane leaves the surface, freeing the sites. A fine powder has a much larger **surface area**, so there are more active sites and the reaction is faster.

6. Michaelis–Menten.

> (a) $[\mathrm{S}] = K_\mathrm{m}$, so $v = \tfrac{1}{2} V_\mathrm{max} = \mathbf{75\ \mu mol\ L^{-1}\ min^{-1}}$
>
> (b)
>
> $$v = \frac{(150\ \mu\mathrm{mol\ L^{-1}\ min^{-1}})(4.5\ \mathrm{mmol/L})}{0.50\ \mathrm{mmol/L} + 4.5\ \mathrm{mmol/L}} = \mathbf{135\ \mu mol\ L^{-1}\ min^{-1}}$$
>
> (c) $0.90 = \dfrac{[\mathrm{S}]}{K_\mathrm{m} + [\mathrm{S}]}$ gives $[\mathrm{S}] = 9K_\mathrm{m} = 9 \times 0.50\ \mathrm{mmol/L} = \mathbf{4.5\ mmol/L}$ (consistent with b).
>
> (d) Exact:
>
> $$v = \frac{(150\ \mu\mathrm{mol\ L^{-1}\ min^{-1}})(0.050\ \mathrm{mmol/L})}{0.55\ \mathrm{mmol/L}} = 14\ \mu\mathrm{mol\ L^{-1}\ min^{-1}}$$
>
> Approximation:
>
> $$v \approx \frac{150\ \mu\mathrm{mol\ L^{-1}\ min^{-1}}}{0.50\ \mathrm{mmol/L}} \times 0.050\ \mathrm{mmol/L} = 15\ \mu\mathrm{mol\ L^{-1}\ min^{-1}}$$
>
> Close (within 10 %), because [S] is well below K_m: the reaction is nearly first order in S.

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D. Challenge

7. Inhibitors.

(a) No inhibitor:

$$v = \frac{(150\ \mu\text{mol L}^{-1}\ \text{min}^{-1})(1.0\ \text{mmol/L})}{1.5\ \text{mmol/L}} = \mathbf{100\ \mu mol\ L^{-1}\ min^{-1}}$$

X: $\dfrac{(150\ \mu\text{mol L}^{-1}\ \text{min}^{-1})(1.0\ \text{mmol/L})}{2.5\ \text{mmol/L}} = \mathbf{60\ \mu mol\ L^{-1}\ min^{-1}}$

Y: $\dfrac{(90\ \mu\text{mol L}^{-1}\ \text{min}^{-1})(1.0\ \text{mmol/L})}{1.5\ \text{mmol/L}} = \mathbf{60\ \mu mol\ L^{-1}\ min^{-1}}$

(b) At 10 mmol/L, X gives about 130 µmol L^{-1} min^{-1} (excess substrate outcompetes it, so the rate approaches 150) but Y gives only about 86 µmol L^{-1} min^{-1} (it can never exceed its lowered V_{max} of 90). The inhibitor whose effect shrinks at high [S] is the competitive one.

8. Turnover and temperature.

(a)

$$k_{cat} = \frac{150\ \mu\text{mol L}^{-1}\ \text{min}^{-1}}{0.025\ \mu\text{mol L}^{-1}} = 6.0 \times 10^{3}\ \text{min}^{-1}$$

$$= \frac{6.0 \times 10^{3}\ \text{min}^{-1}}{60\ \text{s/min}} = \mathbf{1.0 \times 10^{2}\ s^{-1}}$$

(b) At first, raising the temperature gives more collisions with enough energy to react, so the rate rises. Above the optimum (often about 37–45 °C for human enzymes), the protein **denatures**: the weak interactions holding its shape break, the active site no longer fits the substrate, and the rate falls sharply.

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